

Practical No : 08

Aim : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Input :

```
bhavi@LAPTOP-IH23AUG3 MSYS ~  
$ nano lru.c
```

```
GNU nano 8.7  
#include <stdio.h>  
  
int main() {  
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int frame[3], time[3];  
    int i,j,k,least;  
    int faults = 0, hits = 0, counter = 0;  
  
    for(i=0;i<3;i++)  
        frame[i] = -1;  
  
    for(i=0;i<12;i++){  
        int found = 0;  
  
        for(j=0;j<3;j++){  
            if(frame[j] == pages[i]){  
                hits++;  
                counter++;  
                time[j] = counter;  
                found = 1;  
                break;  
            }  
        }  
  
        if(!found){  
            least = 0;  
            for(j=1;j<3;j++){  
                if(time[j] < time[least])  
                    least = j;  
            }  
  
            frame[least] = pages[i];  
            counter++;  
            time[least] = counter;  
            faults++;  
        }  
    }  
  
    printf("Page Hits = %d\n", hits);  
    printf("Page Faults = %d\n", faults);  
    printf("Hit Percentage = %.2f%%\n", (hits/12.0)*100);  
    printf("Fault Percentage = %.2f%%\n", (faults/12.0)*100);  
  
    return 0;  
}
```

Output :

```
bhavi@LAPTOP-IH23AUG3 MSYS ~  
$ ./lru  
Page Hits = 2  
Page Faults = 10  
Hit Percentage = 16.67%  
Fault Percentage = 83.33%
```